

VR Zen Garden: Designing a Virtual Environment for Stress Relief

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Demo Video: youtu.be/n2F5Gs9k41k

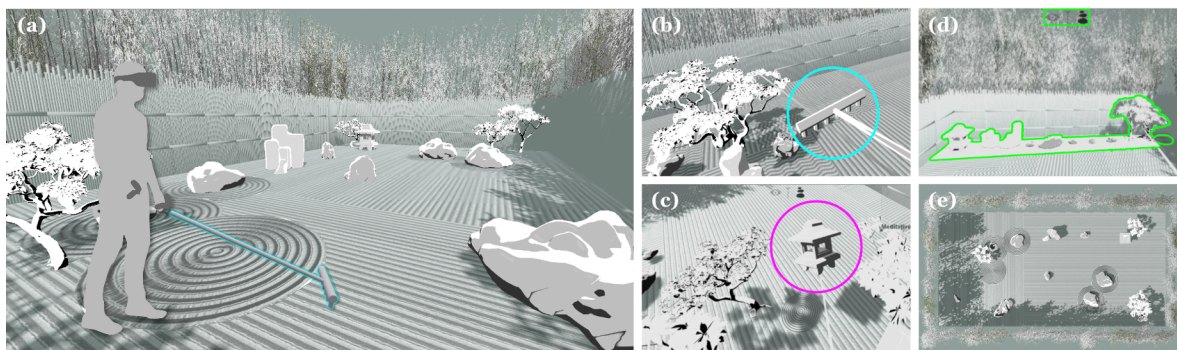


Figure 1: System Overview. (a) Representative view of the VR Zen Garden. (b) Sand pattern raking interaction (cyan). (c) Object placement interaction (magenta). (d) 2D and 3D user interface components (green). (e) Top-down view of the entire VR scene.

ABSTRACT

We propose VR Zen Garden, a virtual reality (VR) application designed to support stress relief and meditation through interactive spatial soundscapes. The system enables users to create personalized auditory environments by combining two key interactions: *Object Placement*, which spatializes sound-emitting objects, and *Sand Pattern Raking*, which modulates sound properties through visual patterns. These inputs are processed by the *Sound Spatializer* and the *Metaphoric Sound Mixer*, allowing users to intuitively shape immersive 3D soundscapes in real time. VR Zen Garden offers an engaging and embodied approach to mindfulness, highlighting the potential of sound-driven design in immersive wellness applications.

Index Terms: Virtual Reality (VR), Stress Relief, Mindfulness, Soundscape

1 INTRODUCTION

Beyond its origins in entertainment, virtual reality (VR) now enables applications in education, therapy, and immersive wellness [5]. Its capacity to craft fully immersive environments supports stress relief and mindfulness practices. Prior research has demonstrated that VR-guided mindfulness interventions can reduce stress levels and facilitate effective relaxation experiences [3, 2].

Despite these advances, limited work has examined how interactive 3D spatial soundscapes and dynamically modifiable auditory

attributes can enhance stress relief in VR. Findings in psychoacoustics highlight the emotional impact of sound: Juslin et al.[4] identified correlations between timbral attributes and emotions. Similarly, Ahonen et al.[1] found that low-frequency stimulation significantly enhanced both physical and emotional relaxation, while reducing stress levels.

To investigate whether spatial auditory cues are effective in inducing emotional relaxation, we propose the VR Zen Garden: an interactive VR system where users design and control personalized 3D environment focusing on soundscapes through spatial object placement and sand pattern interactions. By leveraging intuitive design and responsive audio feedback, the system aims to support immersive, sound-driven stress relief experiences.

2 SYSTEM OVERVIEW

VR Zen Garden is an interactive system that links user actions in a virtual environment with dynamic spatial sound generation. The system combines user-driven interactions with real-time audio processing to create a personalized and immersive auditory experience for stress relief and mindfulness.

Users engage with the system through two core interactions: placing objects that emit spatialized sounds and drawing sand patterns that influence how those sounds are modulated. These interactions are supported by two processing modules: the *Sound Spatializer*, which handles spatial rendering of object-based sounds, and the *Metaphoric Sound Mixer*, which modulates those sounds based on the visual layout of sand patterns.

The system was implemented using Unity 2022.3.24f1¹ and was deployed on the Meta Quest 3 head-mounted display².

Object Placement. Users place sound-emitting objects into the VR Zen Garden using a 3D interface panel. Each object corresponds to a distinct auditory source (e.g., wind, insects, instruments), and the *Sound Spatializer* renders them in 3D space, with attenuation based on their distance and direction relative to the user. This mapping between visual objects and their corresponding auditory sources is

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¹Unity: <https://unity.com/>

²Meta Quest 3: <https://www.meta.com/quest/quest-3/>

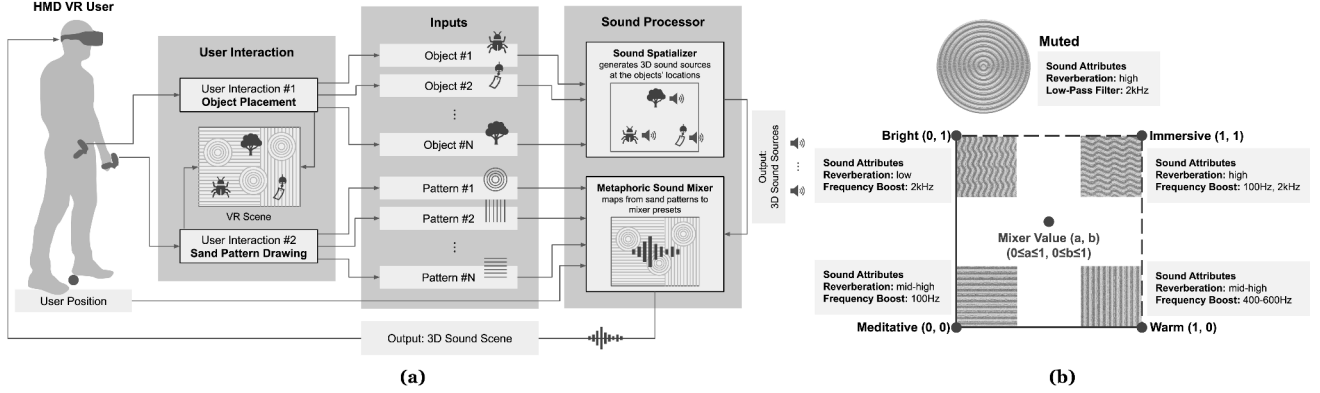


Figure 2: (a) System Diagram of VR Zen Garden. (b) Metaphoric Sound Mixer

illustrated in Figure 3, where each object is assigned a unique sound profile.



Figure 3: Mapping between visual objects and auditory sources. Each placeable object represents a distinct sound-emitting source (e.g., wind, insects, instruments), which the *Sound Spatializer* renders in 3D space.

Sand Pattern Drawing. By sweeping a controller ray across the garden floor, users can draw patterns on a tiled grid ($0.5\text{m} \times 0.5\text{m}$ per cell). Each tile can be assigned one of five visual patterns: *Vertical Wave*, *Vertical Linear*, *Horizontal Linear*, *Horizontal Wave*, and *Swirl*, each corresponding to a specific audio preset: *Bright*, *Warm*, *Meditative*, *Immersive*, and *Muted*, respectively.

Real-time Sound Mixing. The *Metaphoric Sound Mixer* dynamically adjusts the overall soundscape based on the user's location within the pattern grid. Except for the *Muted* state, each preset is encoded as a 2D mixing vector—*Meditative* (0,0), *Bright* (0,1), *Warm* (1,0), *Immersive* (1,1)—and the final sound mix is interpolated from nearby cell values:

$$w_i = \frac{r_{\max} - |p_{\text{user}} - c_i|}{r_{\max}}, \quad (a, b) = \frac{\sum_i w_i \cdot (a, b)_i}{\sum_i w_i} \quad (1)$$

Here, w_i is a spatial weight assigned based on proximity to the user's position p_{user} , and $(a, b)_i$ is the mixing value of each pattern cell i . This ensures that closer patterns have more influence on the auditory output. The resulting value (a, b) determines the mixing characteristics of four primary audio presets, dynamically shaping the 3D soundscape as users move.

3 CONCLUSION

We evaluated the effectiveness of VR Zen Garden in promoting stress relief through a user study with 36 participants, following a procedure adapted from [3]. Participants completed a stress-inducing task and were then assigned to either a traditional meditation session or an interactive session in the VR Zen Garden. Stress levels were assessed before and after the task using the State-Trait Anxiety Inventory (STAI) [6].

Participants who used the VR Zen Garden reported a greater reduction in stress levels (mean decrease of 24.75 points) compared to those in the meditation condition (mean decrease of 18.58 points). These results suggest that immersive, interactive soundscape environments may provide a more effective medium for stress relief in virtual reality settings.

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